

Solution

HUMAN EYE -04

Class 10 - Science

1. It is that defect in the eye due to which a person is unable to distinguish between certain colours due to insufficient or no cone shaped cell on retina. We know there are three types of cone cells - one responding to blue, second to red and third to green colour. The colour blind persons do not possess some cone cells that respond to certain colours. Colour blindness is a genetic disorder caused due to inheritance. There is no cure to this defect. A colour blind person can see everything clearly but is not able to distinguish between different colours.

The founder of atomic theory of matter, Dalton was colour blind.

2. They have very large number of rods on retina. Hence they are able to see even a small quantity of light. In humans, number of rods and number of cones are equal.
3. The eye can focus object at different distances by changing the focal length of its lens. The focal length of the eye lens is changed by the action of ciliary muscles, which can change the thickness of the soft and flexible eye lens such that its focal length (and hence the converging power) is changed.



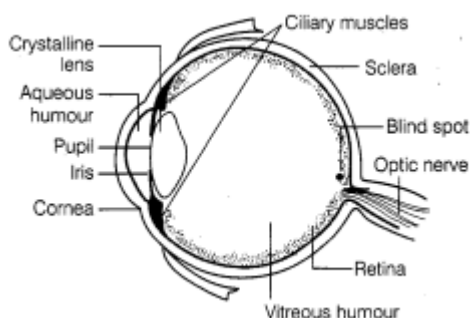
4. It is due to no or insufficient number of cones on the retina. Seeing sun or towards it during solar eclipse may cause colour blindness.
5. The minimum distance, at which objects can be seen most distinctly without strain, is called the least distance of distinct vision. It is also called the near point of the eye. For a young adult with normal vision, the near point is about 25 cm. The farthest point up to which the eye can see objects clearly is called the far point of the eye. It is infinity for a normal eye. A normal eye can see objects clearly that are between 25 cm and infinity.
6. The ability of the eye lens, to adjust its focal length, so as to clearly focus the rays coming from distant as well as near objects on the retina, is called the power of accommodation of the eye. This is accomplished by the ciliary muscle, which controls the eye lens, allowing it to flatten or thicken as is needed for distant or near vision.
7. The eye lens of the human eye is a convex lens that is thick in the middle and thin from the edges. It converges the incident light rays and forms the image on retina.

Role of ciliary muscles:

- i. It changes the shape of the lens in eye to help with focussing.
 - ii. It helps to regulate the flow of aqueous humor in eye.
8. The pupil of the eye acts as a variable aperture whose size can be varied with the help of the iris. When the light is very bright, the iris contracts the pupil to allow less light to enter the eye. However in dim light the iris expands the pupil to allow more light to enter the eye. When we enter from a bright light area to a dim light area, the pupil is not able to expand quickly to allow more light into the eye. We are therefore not able to see the objects in dim light room momentarily.
 9. The light-sensitive cells on retina are activated when the image is formed on it.
The general electrical signals are transmitted to the brain through optic nerve.
The brain, while interpreting and processing these signals, reinvert the image formed and we see the object erect.
In this way, we see objects erect when the eye forms a real, inverted image of them on retina.
 10. The most distant point upto which the eye can see is called the far point. For normal eye, far point is infinity. The point at the shortest distance from the eye upto which the eye can see clearly is known as the near point. It is about 15 cm for a normal eye. The distance between far and near points is called range of vision. Within the range of vision, there is a certain distance where the object is most clearly seen. The distance for a normal eye is about 25 cm and is known as the least distance of distinct vision.
 11. Chickens have very few rods on the retina, hence they are able to see only in intense light and not in dim-light.
 12. Function of
 - i. Iris controls the size of pupil thus, monitoring the amount of light rays entering through pupil.
 - ii. Pupil is the aperture through which light enters and reach retina.

- iii. Cornea provides 67% of eye's focussing power.
13. Accommodation is the distance between near point and far point. The eye is able to adjust its focal length in such a way that it is able to focus any of the objects between near point and far point.
14. a. Function of the following are:
- Retina - The retina is a thin layer of tissue that lines the back of the eye on the inside. It is located near the optic nerve.
 - Pupil - The pupil is a hole located in the centre of the iris of the eye that allows light to strike the retina.
- b. Ciliary muscles are responsible for bringing change in the focal length of the eye lens so that our eye lens can form sharp image of objects at different distances clearly on our retina.
15. The eye lens is composed of fibrous, jelly-like material. To see closure and distant object, its focal length can be changed to a certain extent by contraction and relaxation of ciliary muscles. The ability of the eye lens to adjust its focal length by contracting and relaxing its ciliary muscles is called accommodation. To see closer objects, ciliary muscles, must decrease the focal length of the eye lens. However, the focal length of the eye lens cannot be decreased below a certain minimum limit. To see an object comfortably and distinctly, the distance of an object should be at least 25 cm from the eyes. Due to this, a normal eye is not able to see clearly the objects placed closer than 25 cm. This distance at which objects can be seen is clearly, is called the least distance of distinct vision or the near point of the eye.
16. A human beings has a horizontal field of view of about 150° with one eye and of about 180° with two eyes. Hence two eyes gives us a wider field of view. With one eye the world will appear to us to be a flat one but two eyes open, the world would takes on third dimension of depth. With two eyes, we are able to judge the distance of the object.
17. i. Retina, it is a sensory membrane where image is formed.
 ii. Ciliary muscles, it changes the focal length of the lens.
 iii. Pupil. aperture through which light enters.
 iv. Eye lens, focus light onto the retina.
18. For a person having normal vision, least distance of distinct vision is $d = 25 \text{ cm}$.

$$P = \frac{100}{d} = \frac{100}{25} = 4D$$
19. The two types of light-sensitive cells on retina of an eye are,
- cones:** The cells that respond to colour are cones.
 - rods:** The cells that respond to intensity of light are rods.
20. The ability of eye lens to adjust its focal length with the help of ciliary muscles, it is possible for the eye to form focussed images from the objects that are located at varying distances from the eye. This mechanism is called accommodation.
21. When light falls on retina, it excites rods and cones. The electric pulses produced are conveyed to brain by optic nerve.
22. Had it become solid, its focal length would have been fixed. We would not have been able to focus objects lying at different distances. In short, accommodation of eye would have been reduced to zero.
23. Labelled diagram of human eye



Blind spot is a spot on the retina present at the point of origin of the optic nerve. Photo receptor cells are absent from this region. They are insensitive to light as both rods and cones are absent.

An image formed there is not seen by the eye.

24. Human eyes have power of accommodation. When we have to see distant objects, the ciliary muscles relaxes thus lens become thin and focal length of the lens increases, and we are able to see distant objects. When we have to see nearby objects, the ciliary muscles contract and lens becomes thick. Due to this, the focal length of the lens decreases and the eye is able to see nearby objects.
25. Ciliary muscles help the eye to change its focal length by adjusting its curvature.
26. A person is said to have developed cataract when the eye lens becomes progressively cloudy resulting in blurred vision. The vision of a person having cataract can be restored after getting surgery done on the eye having cataract. The opaque lens is

removed from the eye by surgical operation and a new artificial lens is inserted in its place.

27. Given, $u = -20\text{cm}$, $v = -30\text{cm}$, $P = ?$

Using lens formula,

$$f = \frac{uv}{v-u} = \frac{(-20) \times (-30)}{-30+20} = -60\text{cm}$$

$$\therefore P = \frac{100}{f} = \frac{100}{-60} = -1.67\text{D}$$

28. No, a person may have normal ability of accommodation and yet he may be myopic or hypermetropic. In fact, myopia may arise when length of eyeball is elongated and hypermetropia may arise when length of eyeball gets shortened.

29. When Reena is diagnosed with short sight problem, the better advice to be considered is that she can wear spectacles. But when she is not comfortable wearing spectacles, I would suggest her to fix lens which won't degrade her looks and will make her feel confident.

30. $f = -2.5\text{ m}$

i. Since the focal length is negative, it is a concave lens and it is used in myopia.

ii. Concave

iii. i. excessive curvature of the cornea.

ii. elongation of the eyeball.

iv. $f = -2.5\text{ m}$

The negative sign of focal length indicates that the lens used is concave hence the boy is suffering from myopia (short-sightedness).

$$\text{Power of the lens} = \frac{1}{f(\text{in m})}$$

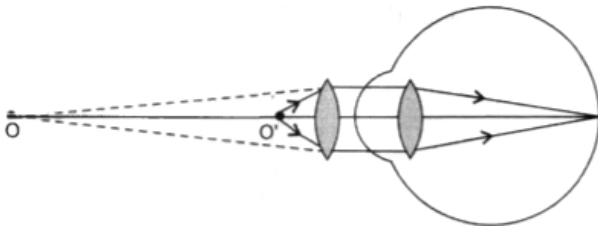
$$= \frac{1}{(-2.5)}$$

$$= -2.5\text{ D}$$

31. Given, far point, $x = 100\text{cm}$

To see the Moon clearly, he should use spectacles with concave lens of focal length, $f = -x = -100\text{cm}$.

32. Hypermetropia can be corrected by using a convex lens of appropriate power. The ray diagram is as follows:



Here,

The object distance, $u = -25\text{ cm} = -0.25\text{ m}$

The image distance, $v = -0.8\text{ m}$

Using the formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

We get

$$\frac{1}{-0.8} - \frac{1}{-0.25} = \frac{1}{f}$$

$$\frac{11}{4} = \frac{1}{f}$$

$$f = \frac{4}{11} = 0.36\text{ m}$$

In order to calculate power,

$$\text{We know, } P = \frac{1}{f(\text{in metres})}$$

$$P = \frac{1}{0.36} = 2.78\text{ D}$$

A convex lens of power $+2.78\text{ D}$ is required to correct the defect.

33. The person is suffering from an eye defect called myopia. In this defect, the image is formed in front of the retina. Hence, a concave lens is used to correct this defect of vision.

Here,

The object distance, $u = \infty$

The image distance, $v = -120\text{ cm} = -1.2\text{ m}$

Using the formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

We get

$$-\frac{1}{1.2} - \frac{1}{-\infty} = \frac{1}{f}$$

$$-\frac{1}{1.2} + 0 = \frac{1}{f}$$

$$f = -1.2\text{m}$$

In order to calculate power,

$$\text{We know } P = \frac{1}{f(\text{ in metres })}$$

$$\text{or } P = \frac{1}{-1.2} = -0.833\text{D}$$

So, a concave lens of power - 0.83 D is required to correct this defect.

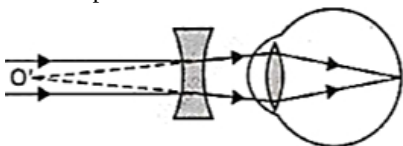
34. Virtual image formed by concave lens:

- a. Diminished
- b. Erect

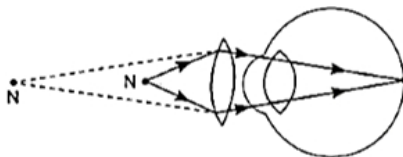
Virtual image formed by convex lens:

- a. Enlarged
- b. Erect

Correction in Myopia: Myopia is also known as near-sightedness. This defect can be corrected by using a concave lens of suitable power. A concave lens of suitable power will bring the image back on to the retina and thus the defect is corrected.



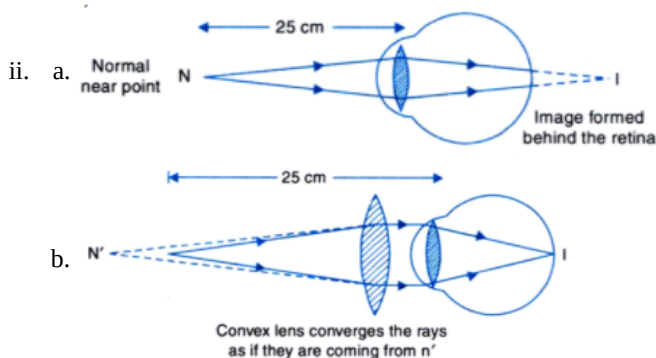
Correction in Hypermetropia: Hypermetropia is also known as far-sightedness. This defect can be corrected by using a convex lens of appropriate power. Eye glasses with converging lenses provide the additional focussing power required for forming the image on the retina.



35. A hypermetropic eye have normal sight for distant objects. He uses spectacles of converging lens to see the nearby objects clearly. If he will use same spectacles to see the distant objects, then parallel rays coming from the distant object may get converged or focussed in front of the retina and so distant objects will appear blurred. So, he has to remove his converging lens spectacles while looking at the distant objects.

36. i. Cause of hypermetropia:

- a. The eyeball becomes too small along its axis so that the distance between the eye lens and the retina is reduced.
- b. The focal length of the eye lens becomes too large resulting in the low converging power of the eye lens.



37. If the spectacle lenses are concave, then the person was short-sighted and if the spectacle lenses are convex, the person was long-sighted.

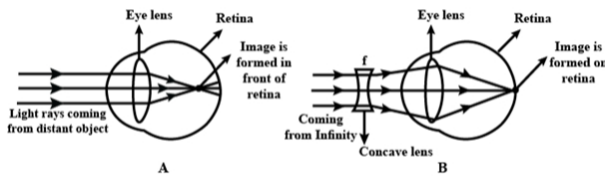
38. Though all the treatment for recovering from his eye problems did not help him out, I would suggest him to opt for eye transplantation which would get his eyesight back to normal. In transplantation of eye, the entire diseased eye cannot be replaced. Only the diseased cornea can be replaced by the donated cornea.

39. Individual is suffering from myopia.

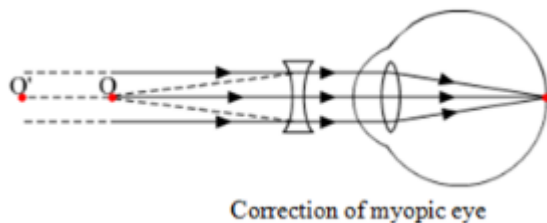
Myopia can be caused by

- a. increase in curvature of the lens.
- b. increase in the length of the eye ball.

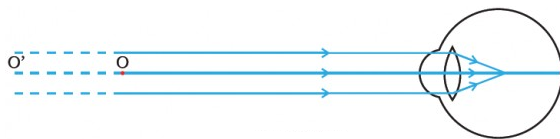
Correction-



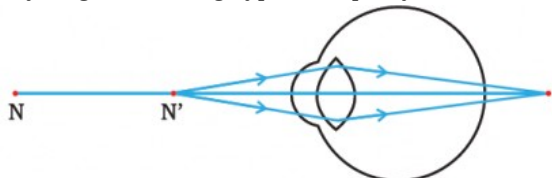
40. i. If the student cannot see the words written on the blackboard or can't see the distant object.
- ii. Causes of myopic eye:
 - a. Excessive curvature of the eyeball
 - b. Elongation of the eyeball
- iii.



- iv. Myopic eye can be corrected by using concave lens.
41. Sometimes, particularly in old age, eyesight becomes foggy. It is due to the eye lens becoming milky and cloudy. When it happens, persons are said to have cataract. There is a loss of vision, sometimes extremely severe. It is possible to treat this defect. The opaque lens is removed and a new artificial lens is inserted. Modern technology has made this procedure simpler and safer.
42. He is suffering from hypermetropia.
43. It is called presbyopia. It can be corrected by using two separate lenses, one for near vision and the other for distant vision.
44. i. Ray diagram showing Myopic eye is:



- ii. Ray diagram showing hypermetropic eye is:



45. For a normal eye, image distance in the eye is fixed, being equal to the distance of the retina from the eye lens. When we increase the distance of an object from the eye, the focal length of the eye lens is changed on account of the accommodation power of the eye, so as to keep the image distance constant.
46. We rarely use concave lens as is usually supposed. We often use convexo concave lens or any other diverging lens.
47. As all digital items and artificial lighting is enriched with blue light which is harmful to our eyes and body, I would suggest Zia to use the digital items in some warmer colour light like yellow or red which does not harm our eye or body when compared with the blue light. Blue light filter application can be installed in computer, mobile or other digital items we use.
48. Two main causes of a person developing near sightedness also known as Myopia are
 - i. Excessive curvature of eye lens
 - ii. Elongation of eyeball
 Can be corrected by using concave lens which is a divergent.

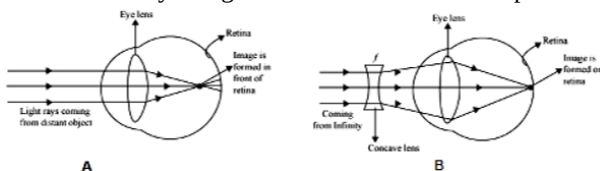
49. $P = \frac{1}{f} = -\frac{1}{1.2} = -0.825D$

Person should wear specs of power -0.825D.

50. i. The defects arise due to the gradual weakening of the ciliary muscles which is used to change the shape of lens in eye to help focussing. Its weakening leads to decreased flexibility of the eye lens. Sometimes, a person may suffer from both myopia and hypermetropia. This condition of defect is called presbyopia. Such people often require bi-focal lenses to correct this defect.
- ii. Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{(-60)} = \frac{1}{f}$
 [given, $v = -60 \text{ cm}$, $u = \infty$]

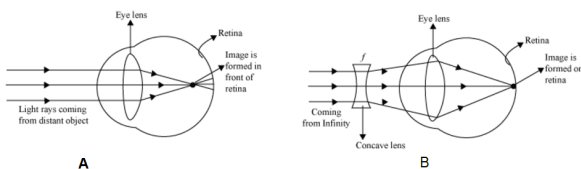
$f = -60$ cm. Thus, the lens used will be concave or diverging lens as the focal length is negative.

51. Though the restriction for mobile and TV has been undergone, still the child faces problem. So, I would suggest my friend for taking care of eating habits of the kid. Eyesight problems can be rectified by eating carrots daily. Either one among the fish, leafy vegetables and egg should be added in daily diet which will decrease the eyesight problems.
52. It was a big problem for Babin to take care of lens in his busy schedule due to this reason I suggested him to undergo laser operation which would correct all common vision problems like near sight, a long sight which reduce the dependency on the lens. The eye treatment using laser is safe and effective.
53. Myopia is a defect of vision in which a person clearly sees all the nearby objects, but is unable to see the distinct object comfortably. A myopic eye has its far point near that infinity. It forms the image of a distinct object in front of the retina as shown in the figure A
- Since concave lens has an ability to diverge incoming rays, it is used to correct this defect of vision. The image is allowed to form at the retina by using a concave lens of suitable power as shown in the figure B.



54. a. Myopia.
b. Concave lens with the focal length of 200 cm
Given, $P = -0.5$ D
We have
 $P = \frac{1}{f}$ m
 $f = \frac{1}{P} = \frac{1}{(-0.5)}$
 $f = -2$ m
 $f = -200$ cm
- c. Two causes of Myopia are:
i. Elongation of eyeball.
ii. High converging power of eye lens.
55. i. He is suffering from myopia (which is also known as short-sightedness or near-sightedness) because the focal length required for the lens is negative.
ii. Given, $P = -1.2$ D
 $\therefore f = \frac{1}{P}$ [Where, $P = \text{power of lens}$ and $f = \text{focal length}$]
 $= \frac{1}{-1.2} = \frac{-10}{12} = -0.83$ m
iii. The lens will be diverging because power is negative, which indicates it is a concave lens.
56. Person is suffering from myopia.
Two possible reasons due to which the defect of vision may have arisen are:
i. increase in curvature of the lens.
ii. increase in length of the eyeball.

Correction-



Lens used is concave lens.

57. Myopia is the nearsightedness and hypermetropia is the far-sightedness. The person who is suffering from both eye defects is advised to wear spectacles with a bi-focal lens which consists of both concave and convex lens. The upper portion would be a concave lens which rectifies the distant vision and lower portion is a convex lens which rectifies the near vision.
58. $\text{power} = \frac{1}{f}$
 $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
object distance = infinite
 $v = -80$

$$-\frac{1}{80} - \frac{1}{\text{infinity}} = \frac{1}{f}$$

$$\frac{1}{\text{infinity}} = 0$$

$$-\frac{1}{80} - 0 = \frac{1}{f}$$

$$f = -80 \text{ cm}$$

$$= -0.8 \text{ m}$$

$$P = \frac{1}{-0.8}$$

$$= -1.25 \text{ D (negative sign indicates that the lens is concave)}$$

A corrective concave lens of power -1.25 D will be required to correct the problem.

59. Here $u = -20 \text{ cm}$, $v = -50 \text{ cm}$, $f = ?$

$$\text{Using } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \text{ we have } \frac{1}{f} = -\frac{1}{50} - \frac{1}{(-20)} = -\frac{1}{50} + \frac{1}{20} = \frac{-2+5}{100} = \frac{3}{100} \text{ or } f = +\frac{100}{3} = 33.3 \text{ cm} = +0.333 \text{ m}$$

$$P = \frac{1}{f} = \frac{1}{+0.333} \text{ or } P = +3 \text{ D}$$

He should use converging lens of power 3D.

60. The surface of the cornea is nearly spherical. Sometimes, its curvature gets distorted and becomes more in one plane than in the other, so that the focal length of the eye in the two planes at right angles to each other becomes different. This defect generally causes indistinct vision and headache and is called astigmatism. This defect is corrected by the use of suitable cylindrical lens.

61. Focal length of convex lens, $f = 5 \text{ cm}$

For the closest distance, $v = -25 \text{ cm}$

$$\text{Using the formula, } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$-\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$$

$$= \frac{1}{5} + \frac{1}{25} = \frac{6}{25}$$

$$u = \frac{-25}{6} \text{ cm} = -4.17 \text{ cm}$$

The closest distance at which Parthiv can read the book is - 4.17 cm.

For the farthest distance, $v' = \infty$

$$\text{Using the formula, } \frac{1}{v'} - \frac{1}{u'} = \frac{1}{f}$$

$$-\frac{1}{u'} = \frac{1}{f} - \frac{1}{v'}$$

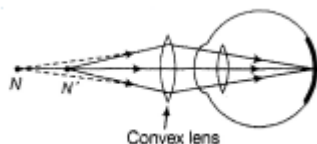
$$= \frac{1}{5} - \frac{1}{\infty} = \frac{1}{5}$$

$$u' = -5 \text{ cm}$$

The farthest distance at which Parthiv can read the book is -5 cm.

62. As a young medico student I would suggest Richie opt for laser cataract surgery in which the hard cloudy substance would be broken very easily by laser instead of microsurgical instrument used in traditional cataract surgery. This laser treatment would be very effective and very precise incisions are made in very less time.

63. i. This defect is hypermetropia (Far-sightedness) as the image of a near point is formed beyond the retina.
 ii. The two possible causes of the defect are (a) Size of eyeball become smaller. (b) The focal length of the lens increases.
 iii. This defect can be corrected by using a convex lens of suitable focal length. The diagram showing correction of this defect by using convex lens of suitable focal length is shown below.



64. Presbyopia is due to decrease in elasticity of eye-lens. With age, the elasticity of eye lens decreases. The near point goes away from the eye and the far point comes near the eye. Due to this, a person suffering from this is not able to see near as well as far away objects clearly. It can be corrected by using bifocal lenses or using two specs, one for reading purpose and one for seeing distant objects.

65. i. a. The type of lens required by such person to improve the vision is bifocal lens.
 b. A bifocal lens consists of both a convex lens and concave lenses. The convex lens used in a bifocal lens is used to correct hypermetropia (farsightedness) and concave lens is used to correct myopia (shortsightedness).

ii. $P = \frac{1}{f}$

$$P_1 = +3 \text{ D} = \frac{1}{f_1}$$

$$F_1 = \frac{1}{3} \text{ m} = +0.33 \text{ m (Convex lens)}$$

$$P_2 = -3D = \frac{1}{f_2}$$

$$F_2 = -\frac{1}{3} = -0.33 \text{ m (Concave lens)}$$

66. The student has difficulty reading the blackboard in the last row implies that the student has difficulty in seeing the distant objects. This type of defect where a person cannot see beyond certain distance is called myopia. It could be corrected by wearing concave lens of suitable power.

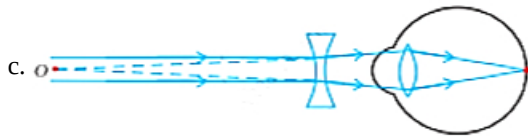
67. Ahmed who is blind to red and green colour may be deficient in cone shaped cells having red and green pigment in the retina of his eyes. It is a genetic disorder and not a refractive defect of vision. So, he may not have normal vision.

68. a. Ray diagram of the defect



Myopic eye

b. In a myopic eye, rather than at the retina itself, the image of a distant object is created in front of the retina. This flaw may result from (i) an excessively curved eye lens or (ii) an elongated eyeball. With the use of a concave lens of the appropriate power, this flaw can be fixed. The flaw is fixed by returning the picture to the retina with a concave lens of the proper power.



Correction for myopic

69. Here $u = -2.5 \text{ m}$; $V = -20 \text{ cm} = -0.2 \text{ m}$; $f = ?$

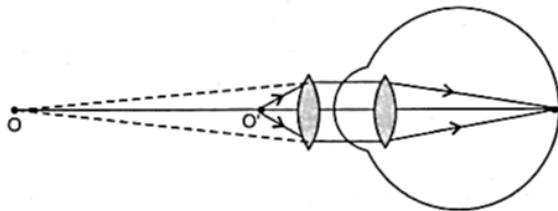
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{(-0.2)} - \frac{1}{(-2.5)}$$

$$\frac{1}{f} = -5 + 0.4 = -4.6 \text{ But } \frac{1}{f} = P$$

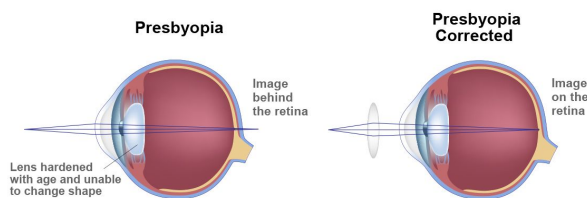
$$\therefore P = -4.6 \text{ D, } f = -\frac{1}{4.6} = -0.2174 \text{ m}$$

He should use concave (diverging) lens of power - 4.6 D.

70. i. The defect of a eye is hypermetropia or long-sightedness.
 ii. The type of lens needed to rectify the above defect is convex lens of appropriate power.
 iii. The diagram showing the rectification of hypermetropia by using a convex lens of suitable power is given below.



71. i. Presbyopia
 ii. This defect occurs in old age due to the gradual weakening of the ciliary muscle and the eye lens becoming inflexible.
 iii. A bifocal lens is often required by persons suffering from this defect.



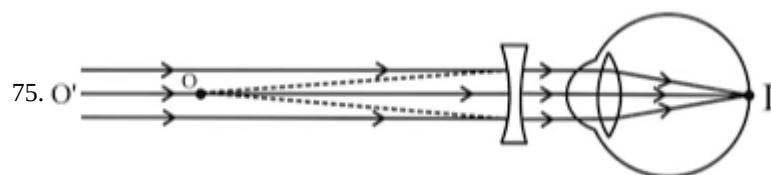
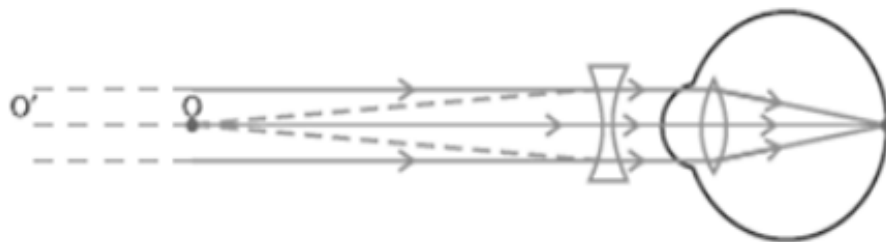
72. i. For distant vision $f_1 = \frac{1}{P_1} = -\frac{1}{5.5} = -0.182 \text{ m}$ or $f_1 = -18.2 \text{ cm}$

ii. For near vision $f_2 = \frac{1}{P_2} = \frac{1}{1.5} = +0.667 \text{ m}$ or $f_2 = 66.7 \text{ cm}$

73. Decrease of Elasticity of Lens. In normal eye, the change in the power of the eye lens for seeing far and near points is very large. As the person grows older, the power of accommodation gradually decreases. A stage may even reach when ciliary muscles lose their power and crystalline lens become much less elastic, so that the power of accommodation is almost zero.

74. The student sitting at the back of the classroom cannot read clearly the letters written on the blackboard, it means that the student is suffering from Myopia. The doctor will suggest to her, a concave lens of suitable power which will bring the image back on the retina, and therefore the defect will be corrected.

Ray diagram for the correction of this defect is as follows:



A bifocal lens is created with two different areas of vision correction, which are divided by a distinct line that sits horizontally across the lens. The top portion of the lens is used for distant, while the bottom portion of the lens is used for closer vision.

A bifocal lens is used for the correction of presbyopia.

Presbyopia is caused by a hardening of the lens of your eye, which occurs with aging.